

Multiple Choice (10 marks)

1. Which of these sets of logic gates are designated as universal gates?
 - (a) XOR, XNOR, NOR.
 - (b) OR, NOT, AND.
 - (c) NOR, NAND, XNOR.
 - (d) AND, OR.
 - (e) OR, XOR, NOT.
2. Find the partial-fraction expansion of $F(s) = [(8s + 2) / (s^2 + 3s + 2)]$
 - (a) $F(s) = [(-2) / (s + 1)] + [(-8) / (s + 2)]$
 - (b) $F(s) = [(8) / (s + 1)] + [(-10) / (s + 2)]$
 - (c) $F(s) = [(8) / (s + 2)] + [(2) / (s + 1)]$
 - (d) $F(s) = [(-8) / (s + 1)] + [(-2) / (s + 2)]$
 - (e) $F(s) = [(6) / (s + 2)] + [(-14) / (s + 1)]$
3. The function f is given by $f(x, y, z) = [z / (x^2 + y^2)]$ (1) Compute in cylindrical coordinates ρ, θ, z .
 - (a) $-(2z / \rho^2)u_\rho + (1 / \rho^3)u_\theta + (1 / \rho)u_z$
 - (b) $(z / \rho^3)u_\rho + (1 / \rho)u_\theta - (1 / \rho^3)u_z$
 - (c) $-(2z / \rho^3)u_\rho + (1 / \rho)u_\theta + (1 / \rho)u_z$
 - (d) $-(z / \rho^2)u_\rho + (1 / \rho^3)u_z$
 - (e) $-(1 / \rho^2)u_\rho - (2z / \rho^3)u_\theta + (2 / \rho)u_z$
4. A 0 - 100 V voltmeter has a guaranteed accuracy of 2 % of full scale reading. The voltage measured by the voltmeter is 75 V. The limiting error is in percentage
 - (a) 1 %.
 - (b) 2.66 %.
 - (c) 2.5 %.
 - (d) 5 %.
 - (e) 2 %.
5. Using the definition of the bilateral z transform $F(z) = \sum_{n=-\infty}^{\infty} f[n]z^{-n}$, find the z-transform of $f[n] = 3\delta[n + 2] + 4\delta[n - 1] + 2\delta[n - 3]$.
 6. $3z^{-2} + 4z^{-1} + 2z^{-3}$

7. $3z^2 + 4z^{-1} + 2z^{-3}$

7. $2z^3 + 4z^{-2} + 3z^{-1}$

7. $4z^2 + 3z^{-1} + 2z^{-3}$

7. $4z^{-3} + 3z^{-2} + 2z^{-1}$

True / False (10 marks)

Answer True or False

6. True or False: The velocity components for the potential function $\phi = x^2 - y^2$ are $V_x = x$, $V_y = -y$, and $V_z = 0$.
7. True or False: The forced current in the given circuit is $i_{\text{forced}} = 30 \cos 1000 t - 15 \sin 1000 t$ mA, which is a common calculation mistake.
8. True or False: A load requiring 60 lb.ft of torque at 1750 r/min can be driven with only 10 hp.
9. True or False: The inverse Laplace transform of $2s / (s^2 + 1)^2$ is $t \cos t$.
10. True or False: The Laplace transform of $f(t) = t^n$ results in $(n+1)! / s^n$, which is incorrect.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Derive the cosine series representation for the piecewise function $f(x)$ defined as $f(x) = 0$ for $0 \leq x < \pi/2$ and $f(x) = 1$ for $\pi/2 < x \leq \pi$, and discuss its significance in engineering applications.
12. Discuss the thermodynamic changes that occur when saturated steam at 100 psia from a steam pipe enters an empty cylinder, and analyze the resulting state of the steam after filling.

End of Paper